WILSON'S INTEGRALITY CONJECTURE FOR RECIPROCAL LCM DETERMINANTS

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Abstract. Let $M_n = (1/\mathrm{lcm}(i,j))_{1\le i,j\le n}$. We prove that $1/\det M_n$ is an integer precisely for $1 \le n \le 34$ and for $n = 36, 38$, resolving the integrality conjecture of Robert G. Wilson V recorded in OEIS A060841. In fact, its 2-adic valuation is negative for every $n \ge 39$. The classical gcd-determinant factorization reduces the problem to the quotient $(n!)^2/\prod_{k=1}^{n}\varphi(k)$. A fixed set of twelve odd primes then gives a uniform negative valuation bound for $n \ge 120$; the remaining cases are settled by exact arithmetic. The proof uses no estimate for the distribution of primes. A self-contained Lean 4 formalization, importing only Mathlib, proves the exact integrality statement in Google DeepMind's Formal Conjectures library, together with the stronger valuation assertion.

1. Introduction

For a positive integer n, consider the rational matrix

$$M_n = \left(\frac{1}{\mathrm{lcm}(i,j)}\right)_{1\le i,j\le n}, \qquad R_n = (\det M_n)^{-1}. \tag{1}$$

The matrix is nonsingular; a standard factorization, recalled below, shows that its determinant is positive. The sequence of numerators of R_n in lowest terms is OEIS A060841. In a comment dated August 2, 2015, Robert G. Wilson V conjectured that the integral values occur exactly at

$$\mathcal{I} = \{1, 2, \ldots, 34, 36, 38\}. \tag{2}$$

The first exception is $n = 35$, already noted by Harry J. Smith in the OEIS entry [3]. We prove the complete classification.

Theorem 1.1. *For every positive integer n,*

$$R_n \in \mathbb{Z} \quad \Longleftrightarrow \quad n \in \mathcal{I}.$$

Moreover, $v_2(R_n) < 0$ *for every* $n \ge 39$.

Here, for a prime q and a nonzero rational number a/b, we use $v_q(a/b) = v_q(a) - v_q(b)$. A negative valuation therefore certifies nonintegrality. The second assertion says that every reduced denominator from $n = 39$ onward is even.

The proof begins with Smith's classical gcd-determinant identity [4], which gives

$$R_n = \frac{(n!)^2}{\prod_{k=1}^{n}\varphi(k)}.$$

Our main observation is that the contributions of only twelve fixed odd primes force $v_2(R_n) < 0$ throughout the infinite range $n \ge 120$. The relevant weighted reciprocal sum is larger than $5/3$, whereas the positive part of the valuation is at most $3n/2 - 1$. This leaves only 119 cases. We give their complete 2-adic certificate, and an elementary argument treats all odd primes at the candidate integral indices.

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Wilson's OEIS comment also proposed that every reduced denominator of R_n is a power of two. That separate assertion is false: the work of Adamczewski [1, Appendix A, entry A060841] records a counterexample at $n = 1807$, for which $v_3(R_n) = -1$. Theorem 1.1 settles the integrality classification, the other assertion in Wilson's comment. In the version of Google DeepMind's Formal Conjectures library consulted on October 3, 2026, the classification is the statement `OeisA60841.conjecture1`, while the denominator assertion is recorded separately [2].

2. The determinant and its valuations

We write φ for Euler's totient function, with $\varphi(1) = 1$. All sums and products indexed by p or q below are over primes unless otherwise specified.

Lemma 2.1. *For every $n \geq 1$,*

$$\det M_n = \frac{\prod_{k=1}^{n} \varphi(k)}{(n!)^2}, \qquad R_n = \frac{(n!)^2}{\prod_{k=1}^{n} \varphi(k)} = n! \prod_{p \leq n} \left(\frac{p}{p-1}\right)^{\lfloor n/p \rfloor}. \tag{3}$$

Proof. Let Z be the $n \times n$ divisibility matrix, defined by $Z_{id} = 1$ if $d \mid i$ and $Z_{id} = 0$ otherwise. Since Z is lower triangular with diagonal entries equal to one, $\det Z = 1$. The identity $\sum_{d \mid m} \varphi(d) = m$ gives

$$G_n := \big(\gcd(i,j)\big)_{1 \leq i,j \leq n} = Z \operatorname{diag}\big(\varphi(1), \ldots, \varphi(n)\big) Z^{\mathsf{T}}. \tag{4}$$

Indeed, the (i,j) entry on the right is the sum of $\varphi(d)$ over the common divisors of i and j.

Set $D = \operatorname{diag}(1, \ldots, n)$. The relation $\gcd(i,j)\operatorname{lcm}(i,j) = ij$ implies $M_n = D^{-1} G_n D^{-1}$. Taking determinants in (4) proves the first identity in (3), including positivity. Finally,

$$\prod_{k=1}^{n} \varphi(k) = n! \prod_{p \leq n} \left(\frac{p-1}{p}\right)^{\lfloor n/p \rfloor},$$

because precisely $\lfloor n/p \rfloor$ integers between 1 and n are divisible by p. Substitution gives the last identity. □

Proposition 2.2. *For every prime q and every positive integer n,*

$$v_q(R_n) = v_q(n!) + \lfloor n/q \rfloor - \sum_{p \leq n} v_q(p-1)\, \lfloor n/p \rfloor. \tag{5}$$

In particular, on writing $E(n) = v_2(R_n)$,

$$E(n) = \sum_{j \geq 1} \left\lfloor n/2^j \right\rfloor + \lfloor n/2 \rfloor - \sum_{3 \leq p \leq n} v_2(p-1)\, \lfloor n/p \rfloor. \tag{6}$$

Proof. Apply v_q to the prime product in (3). The numerator p contributes only when $p = q$. For $q = 2$, the term $p = 2$ in the negative sum vanishes, and Legendre's formula gives $v_2(n!) = \sum_{j \geq 1} \left\lfloor n/2^j \right\rfloor$. □

3. A finite set of primes controls the infinite range

The following elementary criterion isolates the part of the argument that is uniform in n.

Lemma 3.1. *Let $\mathcal{P}$ be a finite set of odd primes, and put*

$$A_{\mathcal{P}} = \sum_{p \in \mathcal{P}} v_2(p-1), \qquad C_{\mathcal{P}} = \sum_{p \in \mathcal{P}} \frac{v_2(p-1)}{p}.$$

Then, for every $n \geq 1$,

$$E(n) \leq \left(\frac{3}{2} - C_{\mathcal{P}}\right) n + A_{\mathcal{P}} - 1. \tag{7}$$

Proof. Legendre's formula gives $v_2(n!) < n$, and hence $v_2(n!) \le n-1$. Terms with $p > n$ contribute zero to $\lfloor n/p \rfloor$, so we may retain all primes in $\mathcal{P}$ in the negative sum of (6). Using $\lfloor n/p \rfloor \ge n/p - 1$ and $\lfloor n/2 \rfloor \le n/2$, we obtain

$$\begin{aligned} E(n) &\le n - 1 + \frac{n}{2} - \sum_{p \in \mathcal{P}} v_2(p-1)\, \lfloor n/p \rfloor \\ &\le n - 1 + \frac{n}{2} - C_{\mathcal{P}} n + A_{\mathcal{P}}, \end{aligned}$$

as required. □

Proposition 3.2. *For every positive integer n,*

$$E(n) < 20 - \frac{n}{6}. \tag{8}$$

Consequently, $E(n) < 0$ for every $n \ge 120$.

Proof. Choose

$$\mathcal{P} = \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41\}.$$

The weights $e_p = v_2(p-1)$ and a small integer certificate for their weighted reciprocal sum are as follows.

p	3	5	7	11	13	17	19	23	29	31	37	41
e_p	1	2	1	1	2	4	1	1	2	1	2	3
$\lfloor 600 e_p/p \rfloor$	200	240	85	54	92	141	31	26	41	19	32	43

Thus $A_{\mathcal{P}} = 21$. The last row sums to 1004, which proves

$$C_{\mathcal{P}} = \frac{255691624204897}{152125131763605} \ge \frac{1004}{600} > \frac{5}{3}. \tag{9}$$

Substituting into (7), and using $n > 0$, yields

$$E(n) < \left(\frac{3}{2} - \frac{5}{3}\right) n + 20 = 20 - \frac{n}{6}.$$

□

4. The finite range and the proof of the theorem

The values of $E(n)$ for $1 \le n \le 119$ are listed in Appendix A. They can be obtained either from (6) or from the recurrence

$$E(1) = 0, \qquad E(n) = E(n-1) + 2v_2(n) - v_2(\varphi(n)) \quad (n \ge 2), \tag{10}$$

which follows immediately from (3). For these computations one may use the formula

$$\varphi(n) = \prod_{p^a \| n} p^{a-1}(p-1),$$

with the empty product equal to one. In this finite range, $E(n) \ge 0$ precisely when $n \in \mathcal{I}$. Nonnegative 2-adic valuation alone does not establish integrality; the next lemma deals with the odd primes.

Lemma 4.1. *If $1 \le n \le 38$ and q is an odd prime, then $v_q(R_n) \ge 0$.*

Proof. The only odd prime divisors of $p-1$ for primes $p \le 38$ are $3, 5, 7, 11$. For every other odd prime q, the negative sum in (5) is zero.

For $q = 5$, that sum is $\lfloor n/11 \rfloor + \lfloor n/31 \rfloor \le 2\lfloor n/5 \rfloor$. Since $v_5(n!) \ge \lfloor n/5 \rfloor$, equation (5) is nonnegative. For $q = 7$ the negative sum is $\lfloor n/29 \rfloor \le \lfloor n/7 \rfloor$, and for $q = 11$ it is $\lfloor n/23 \rfloor \le \lfloor n/11 \rfloor$. These inequalities also suffice.

For $q = 3$, write the negative sum as

$$T(n) = \lfloor n/7 \rfloor + \lfloor n/13 \rfloor + 2\lfloor n/19 \rfloor + \lfloor n/31 \rfloor + 2\lfloor n/37 \rfloor.$$

An exact calculation gives

$$\frac{1}{7}+\frac{1}{13}+\frac{2}{19}+\frac{1}{31}+\frac{2}{37}=\frac{815785}{1983163}<\frac{1}{2},$$

so $T(n)<n/2$. If $n\geq 8$, then

$$\begin{aligned} v_3(R_n) &\geq 2\lfloor n/3\rfloor - T(n) \\ &> \frac{2(n-2)}{3}-\frac{n}{2}=\frac{n-8}{6}\geq 0. \end{aligned}$$

For $1\leq n\leq 6$, $T(n)=0$. Finally, $v_3(R_7)=v_3(7!)+\lfloor 7/3\rfloor -1=3$. □

Proof of Theorem 1.1. Proposition 3.2 gives $E(n)<0$ for $n\geq 120$. The finite certificate gives the same inequality for $39\leq n\leq 119$ and for $n=35,37$. Thus every $n\notin\mathcal{I}$ is excluded by its 2-adic valuation, and the stronger assertion for $n\geq 39$ follows.

If $n\in\mathcal{I}$, the certificate gives $v_2(R_n)\geq 0$ and Lemma 4.1 gives $v_q(R_n)\geq 0$ for every odd prime q. A positive rational number is an integer if and only if all of its prime valuations are nonnegative. Hence $R_n\in\mathbb{Z}$ for each $n\in\mathcal{I}$. □

5. Formal verification and reproducibility

The companion file

`Wilson_Integrality_Conjecture_A060841.lean`

contains a complete Lean 4 proof and imports only Mathlib. It is a single source file; no auxiliary proof files or external computational certificates are required. It reproduces the definitions `OeisA60841.lcmMatrix` and `OeisA60841.integerDetN` from the Formal Conjectures source [2].

Its final theorem, `OeisA60841.conjecture1`, has exactly the type of the repository's integrality statement: for $n\geq 1$, the reduced denominator of $(\det M_n)^{-1}$ equals one if and only if $n\in\{1,\dots,34,36,38\}$. The stronger assertion $v_2(R_n)<0$ for $n\geq 39$ is also proved, as `WilsonA060841.two_adic_negative`.

The formal argument derives the determinant formula directly from the original matrix definition. It then proves the fixed-prime valuation bound uniformly for $n\geq 120$. The finite integrality cases are checked by `decide +kernel` on the exact rational expression $(n!)^2/\prod_{k=1}^{n}\varphi(k)$; the finite negative-valuation cases are checked in the same way. Thus the main classification does not depend on importing an unproved conjecture or on accepting a table as an axiom. The hand proof in Lemma 4.1 provides a separate explanation of integrality at the exceptional indices.

The file was checked with Lean 4.33.1 and Mathlib revision

`0df444a360eaa60ab8c11dca51a86af692955474`.

Both main declarations have the axiom audit

`[propext, Classical.choice, Quot.sound]`.

They use only these standard Lean axioms, with no `sorry`, additional axioms, or `native_decide`. In a Lean project using that Mathlib revision, the source is checked with

`lake env lean Wilson_Integrality_Conjecture_A060841.lean`.

As supplementary numerical checks, the development scripts compute all 119 rational products, compare them with the leading principal determinants obtained by exact rational elimination, and repeat the finite classification using integer factorization and divisibility. These computations corroborate the certificate; the Lean proof checks the finite range in its kernel and proves the infinite range symbolically.

Appendix A. Complete finite valuation certificate

Each row of Table 1 lists the values of $E(n)$ at successive integers in the indicated interval. In particular, $E(35)=-1$, $E(36)=1$, $E(37)=-1$, and $E(38)=0$. The recurrence (10), together with the elementary factorization of $\varphi(n)$, gives a short exact procedure for reproducing every entry.

TABLE 1. The values $E(n) = v_2(R_n)$ for $1 \le n \le 119$.

n	$E(n)$, in increasing order of n									
1–10	0	2	1	4	2	3	2	6	5	5
11–20	4	6	4	5	2	7	3	4	3	4
21–30	2	3	2	5	3	3	2	4	2	1
31–40	0	6	4	2	-1	1	-1	0	-3	-1
41–50	-4	-4	-5	-3	-6	-5	-6	-2	-3	-3
51–60	-8	-7	-9	-8	-11	-8	-10	-10	-11	-11
61–70	-13	-12	-14	-7	-11	-11	-12	-13	-15	-16
71–80	-17	-14	-17	-17	-20	-18	-20	-21	-22	-19
81–90	-20	-21	-22	-21	-27	-26	-29	-26	-29	-30
91–100	-33	-31	-33	-32	-35	-30	-35	-34	-36	-35
101–110	-37	-40	-41	-39	-43	-43	-44	-42	-44	-45
111–119	-48	-44	-48	-48	-51	-50	-53	-52	-57	

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